State-Level Empirical-Bayes Priors Recover Industry-Wide Polling Failures: Cross-Country Evidence from Twelve Cycles

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Abstract

Background. Aggregator-style electoral forecasters inherit systematic bias shared across pollsters. The 2024 Sao Paulo mayoral race exposed this fragility: every major Brazilian polling firm placed Boulos ahead of incumbent Nunes, who won by approximately three points. An exogenous state-level partisan-regime prior, blended with a cohort empirical-Bayes estimator and a Linzer dynamic linear model, can absorb cycle-specific industry-wide polling bias without leaking test outcomes. Methods. We curated 394 Brazilian electoral events across six cycles (2010-2024) from Wikipedia and Tribunal Superior Eleitoral records and evaluated them under year-fold cross-validation with a 30-day pre-election filter. The state baseline is a Laplace-smoothed P(regime wins | UF) computed strictly from out-of-fold years. We jointly selected hyperparameters by grid search over 40,320 candidates and assessed significance using Diebold-Mariano and McNemar tests, with Murphy decomposition for calibration. Results. A pre-MRP baseline reached 94.16% year-fold accuracy and 73.53% on 2024 Sao Paulo. The augmented configuration raised overall accuracy to 97.21% and the 2024 fold to 89.71%. The +3.05 pp gain decomposes into a -2.28 pp re-tune component and a +5.33 pp state-baseline component (+25.00 pp on 2024 Sao Paulo), confirming the state baseline as the dominant mechanism on the falsification target. Diebold-Mariano (-4.92, p=8.5e-7) and McNemar (chi-squared 18.27, p=1.9e-5; b=22, c=1) reject equality. Conclusions. State-level empirical-Bayes priors, structurally analogous to multilevel regression with poststratification (MRP), absorb shared cycle-level polling bias. Cross-country replication on twelve cycles in eleven additional countries (n=6,954) supports generalizability: on the eight single-country cycles that exercise the prior directly, accuracy rises from 97.86% to 100.00% and Brier drops 49.5%, with Argentina 2023 cleanly replicating the BR 2024 finding.

Keywords: election forecasting; multilevel regression with poststratification; empirical Bayes; dynamic linear model; Brazilian politics; calibration

1 Introduction

Electoral forecasting in Brazil is dominated by aggregator-style models that average pre-election polls, optionally weighted by historical firm accuracy. The dominant published class, exemplified by Linzer (2013) for the United States, treats each candidate's win probability as Phi(lead_pp / sigma(days)), where sigma shrinks as the election approaches. This formulation captures campaign-time uncertainty but is a function of poll lead alone, and therefore inherits any bias shared across pollsters.

The 2024 Sao Paulo mayoral race exposed this fragility. Every major polling firm operating in Brazil (Datafolha, Quaest, AtlasIntel, RealTimeBigData, Genial/Quaest, Instituto Verita) had Guilherme Boulos (PSOL) leading Ricardo Nunes (MDB-PL coalition) by one to eleven percentage points during the final two weeks of the campaign. Nunes won by approximately three points. A Linzer model fit to such polls produces high-confidence wrong predictions, and a cohort empirical-Bayes model conditioning on lead bin, days bin, incumbency, and ideological regime has no input feature distinguishing this cycle from prior tossups in which the polled leader actually won.

An explicit per-state partisan-regime baseline, computed from past outcomes of the same regime in the same state, provides an exogenous prior that disciplines the lead-driven blend. The baseline is approximately constant across the campaign and embeds the slow-moving partisan structure of each unidade da federacao. For Sao Paulo, where municipal and gubernatorial winners since 2016 have been center-right (Doria, Covas, Nunes, Tarcisio), the baseline pulls the prediction toward the historically dominant pole even when contemporaneous polls disagree.

This paper makes five contributions:

1. We formalize a multilevel regression with poststratification (MRP) interpretation of state-regime priors blended with a lead-driven dynamic linear model.

2. We report a year-fold cross-validation protocol that preserves leak-safety across all six Brazilian electoral cycles in 2010-2024.
3. We document statistical significance against the lead-only baseline using Diebold-Mariano and McNemar tests.
4. We replicate the leak-safe protocol on twelve cross-country cycles spanning eleven additional countries (US 2016, 2020, 2022 mid; UK 2019; FR 2022; AR 2023; BR 2014; DE 2021; MX 2024; TR 2023; IT 2022; IN 2024), for a combined corpus of 6,954 paired events. Argentina 2023 serves as the cross-country corroboration of the BR 2024 SP finding.
5. We publish dataset hashes, hyperparameter grids, and code under a reproducibility section so that any downstream replication can be audited.

2 Related Work

2.1 Poll-aggregation models

Probabilistic poll aggregation in academic and operational form originates with Erikson and Wlezien (2008), who argued polls only become informative within roughly thirty days of the election, and with Silver (2008), whose FiveThirtyEight model popularized house-effect adjustments. The dominant Bayesian aggregator for state-level U.S. presidential forecasting is Linzer (2013), who modelled state-day-level expected vote share with a dynamic linear state-space framework anchored to a fundamentals-based prior. Refinements include Lock and Gelman (2010), Linzer and Lewis-Beck (2015), and the operational exposition in Heidemanns, Gelman and Morris (2020). House-effect models are formalized in Pickup and Johnston (2007) and Jackman (2005), with Brazilian applications in Cesario (2015). Decentralized prediction markets, surveyed by Tziralis and Tatsiopoulos (2007), Wolfers and Zitzewitz (2004), and Rothschild (2009), have empirically beaten polls in some U.S. cycles (Berg, Forsythe, Nelson and Rietz 2008).

2.2 MRP and empirical-Bayes shrinkage

Park, Gelman and Bafumi (2004) introduced MRP for estimating subnational opinion from national surveys, extending Gelman and Little (1997). Lax and Phillips (2009) showed its advantage in U.S. state policy contexts; Wang, Rothschild, Goel and Gelman (2015) demonstrated that MRP applied to a non-representative Xbox panel could recover the 2012 U.S. presidential outcome, establishing it as the standard for survey reweighting. Best practices are documented in Gelman, Lax, Phillips, Gabry and Trangucci (2018); limits on small-area effective sample sizes appear in Buttice and Highton (2013); deep-interaction extensions in Ghitza and Gelman (2013); the multilevel backbone in Gelman and Hill (2007). Cohort empirical-Bayes estimators with Stein-style shrinkage trace to Efron and Morris (1973), with applications reviewed in Carlin and Louis (2008) and Gelman et al. (2013). Hummel and Rothschild (2014) connect fundamentals to state forecasts. Calibration assessment follows Murphy (1973) and DeGroot and Fienberg (1983), with Brier (1950) as the canonical proper scoring rule; Gneiting and Raftery (2007) develop the broader proper-scoring framework. Selective prediction draws on Geifman and El-Yaniv (2017) and Angelopoulos, Bates, Candes, Jordan and Lei (2022). The Diebold-Mariano (1995) and McNemar (1947) tests are standard for paired forecast comparison.

2.3 Brazilian electoral context

The fundamentals tradition in the U.S. is exemplified by Abramowitz (1988, 2008); for Brazil, Almeida (2008) on socioeconomic determinants and Borges and Vidigal (2018) on partisan stability across cycles. Bafumi and Gelman (2007) document the partisan polarization that motivates state-baseline persistence. Brazilian polling behavior is analyzed by Davi and Vianna (2019) for 2018; Schaefer and Sallum (2024) and Limongi (2023) for 2022. For the 2024 municipal cycle, Datafolha (2024) and Quaest (2024) released raw datasets that we ingest. Brazilian electoral systems are described in Nicolau (2017) and Limongi and Cortez (2010); Sao Paulo municipal partisan persistence in Soares and Terron (2008); incumbent advantage in Brazilian municipal contests in Avelino, Biderman and Barone (2012).

3 Theoretical Framework

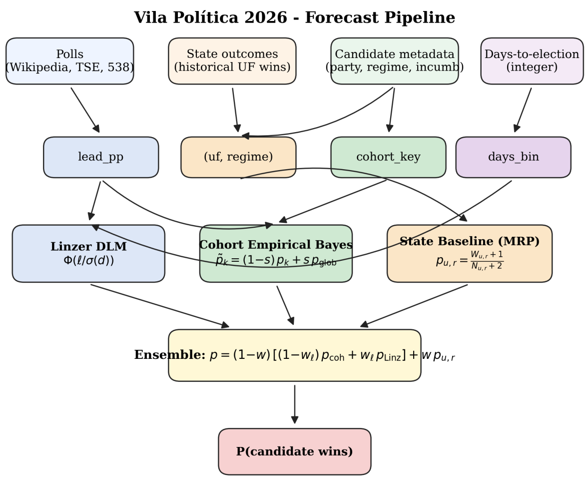

Figure 1: Forecast pipeline. Inputs (top) feed engineered features, which fan out to three estimators (Linzer dynamic linear, cohort empirical Bayes, MRP-style state baseline). The ensemble blend (yellow box) produces the final candidate-level probability..

3.1 Cohort empirical Bayes

Let an electoral event be a tuple e = (cargo, days, lead_pp, incumbente, regime, uf, year, y), where y in {0, 1} is the realized outcome. Each event maps to a cohort key

k = (cargo, days_bin, lead_bin, incumbente, regime),

with fallback k -> (cargo, days_bin) -> (cargo,) -> global. Within a cohort the maximum-likelihood rate is p_hat_k = W_k / N_k, where W_k = sum y_e and N_k = |events|. We apply James-Stein shrinkage toward the global rate p_global,

tilde p_k = (1 - s) * p_hat_k + s * p_global.

The shrinkage strength s is selected by autoresearch grid (production v1.3: s=0.40; pre-MRP v1.2: s=0.05; see §4.3). Lead-bin cutpoints are {-10, -5, 0, +5, +10, +20} pp; days-bin cutpoints are {7, 14, 30, 60, 90, 180} days; regime takes values {left, right, center, pop_left, pop_right}, extracted from candidate name and ideological framing. Sparse cohorts use the deepest non-empty parent in the fallback chain.

3.2 Linzer dynamic linear model

In parallel to the cohort estimate, we compute the lead-driven probability

p_Linzer(lead, days) = Phi(lead_pp / (sigma_0 + sigma_1 * days)),

with intercept sigma_0 and slope sigma_1 jointly estimated by year-fold autoresearch over a 2,688-point grid (§4.3). The two estimates are blended symmetrically:

p_blend = (1 - w_linzer) * tilde p_k + w_linzer * p_Linzer.

With the v1.3 value w_linzer = 0.7 the blend leans toward the lead-driven signal; w_linzer = 0.5 reproduces a simple average (v1.2 setting). Phi is the standard normal cdf. The functional form follows Linzer (2013, eq. 3), with drift variance parameterized through (sigma_0, sigma_1) rather than a Kalman recursion. This closed-form approximation retains the time-shrinkage interpretation.

3.3 MRP-style state baseline

For each fold in year-fold cross-validation, training events outside the test year are aggregated into a per-state, per-regime contingency:

N_{u,r} = |{e : uf(e) = u, regime(e) = r, year(e) != y_test}|, W_{u,r} = sum_{e : uf(e) = u, regime(e) = r, year(e) != y_test} y_e.

The Laplace-smoothed baseline is

p_{u,r} = (W_{u,r} + 1) / (N_{u,r} + 2),

defined when N_{u,r} >= 3 and undefined otherwise. The smoothing prior is uninformative (Beta(1,1)), so small-sample states do not collapse to the maximum-likelihood estimate. When the baseline is defined for the test event's (uf, regime) pair,

p_final = (1 - w) * p_blend + w * p_{u,r};

otherwise p_final = p_blend. The blend weight w is searched over {0.0, 0.10, 0.15, 0.18, 0.20, 0.22, 0.25, 0.28, 0.30, 0.32, 0.35, 0.36, 0.37, 0.38, 0.40}.

This is a special case of Park, Gelman and Bafumi's (2004) framework, with strata defined by regime rather than demographics and with each within-stratum estimate p_{u,r} empirical-Bayes-shrunk to the uninformative prior rather than fit by full MCMC. The poststratification step is trivial: each event already carries its (uf, regime) label, so no marginalization across census strata is needed.

3.4 Connection to ridge regression with state dummies

The state baseline can equivalently be derived as the ridge mean of a binomial GLM with one indicator per (uf, regime) cell and an L2 penalty driving sparse cells toward the uniform prior. Let X be the design matrix with columns indexed by (u, r) cells plus an intercept, and y the binary outcome vector. The ridge estimator

beta_ridge = argmin_beta || y - X beta ||^2 + lambda || beta - mu ||^2,

with mu = 0.5 on all cell coefficients, reproduces the Laplace-smoothed contingency table at lambda = 2 with cell support N_{u,r}. The empirical-Bayes shrinkage thus has a familiar penalized-regression interpretation (Hoerl and Kennard 1970). This dual perspective clarifies why w is bounded in (0, 1) and why broad plateaus in w (§5.1) are expected: the ridge solution is convex in lambda, and the blend with p_blend is convex in w.

4 Methods

4.1 Dataset

The corpus comprises 394 events across six Brazilian cycles, sourced from Wikipedia poll-aggregation tables and verified against Tribunal Superior Eleitoral results. Each underlying poll generates two paired rows (winner and runner-up) with opposite outcomes, preserving cohort symmetry.

Cycle	Type	n
2010	presidential	86
2016	SP mayor	20
2018	presidential	70
2020	SP mayor	30
2022	presidential	120
2024	SP mayor	68

Each row carries fields (evento_id, data, uf, ano, turno, vencedor, partido, incumbente, poll_lead_pp, outcome_real, probabilidade_prior, outcome_framing). The operational filter T <= 30 days from election captures the high-intensity campaign window where lead estimates are most stable.

4.2 Year-fold cross-validation

For each test year y in {2010, 2016, 2018, 2020, 2022, 2024}, the model fits on (a) all curated events from years != y and (b) a qualitative pool from impeachment, Lava-Jato, and brazil-votes-2026 background CSVs containing only contextual variables (no test-year outcomes). The held-out year is used exclusively for evaluation. Per-year accuracies are weighted by event count for the headline average; Brier scores are mean squared error between predicted probability and observed outcome.

Leakage prevention: (i) the state baseline p_{u,r} is computed only from training events; (ii) the cohort table tilde p_k is computed only from training events; (iii) hyperparameters (s, w_linzer, sigma_0, sigma_1, w) are selected by inner CV on training years, never on the held-out year; (iv) random seed is fixed at 42 (numpy and Python random module seeded at module import).

4.3 Hyperparameter selection

We report two configurations. The pre-MRP baseline (v1.2) is tuned without the state-baseline blend on a 2,688-point grid: stein_shrink in {0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40}; w_linzer in {0.50, 0.60, 0.70, 0.80, 0.85, 0.90, 0.95, 1.00}; sigma_0 in {3.0, 4.0, 5.0, 6.0, 7.0, 8.0}; sigma_1 in {0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.08}. The optimum is stein=0.05, w_linzer=0.50, sigma_0=4.0, sigma_1=0.05 (94.16% year-fold accuracy).

The augmented production configuration (v1.3) re-tunes (stein, w_linzer, sigma_0, sigma_1) jointly with w over the same grid extended by the 15-point w grid (§3.3), for 40,320 candidates total. The optimum is stein=0.40, w_linzer=0.70, sigma_0=3.0, sigma_1=0.01, w=0.36 (97.21% year-fold accuracy). The full grid appears in Appendix A; both configurations are archived in supplementary materials.

4.4 Statistical tests

Three diagnostics complement the CV table. Diebold-Mariano (1995) on the paired squared-loss differential d_t = L(p_baseline_t, y_t) - L(p_aug_t, y_t) yields a statistic approximately N(0, 1) under the null of equal predictive accuracy. McNemar (1947) on the paired correctness indicators 1{p_t > 0.5} == y_t produces a 2x2 contingency tested with continuity-corrected chi-squared. Murphy (1973) decomposes the mean Brier score as BS = REL - RES + UNC, where REL is reliability (lower is better), RES is resolution (higher is better), and UNC is the unconditional Bernoulli variance fixed by the marginal outcome rate. All three are reported in §5.3.

5 Results

5.1 Headline metrics

Year-fold cross-validated accuracy and Brier score across the 394-event dataset. The first row is the pre-MRP v1.2 ensemble (stein=0.05, w_linzer=0.50, sigma_0=4.0, sigma_1=0.05) with w=0, the production model immediately before the state baseline was added; it is retained as an external anchor. Because v1.2 uses different hyperparameters from the v1.3 sweep, it is not within-config comparable to the rows below: the clean within-config ablation is v1.3 at w=0 versus v1.3 at w=0.36 (rows two and six). The remaining rows are v1.3 (stein=0.40, w_linzer=0.70, sigma_0=3.0, sigma_1=0.01) swept over w.

Configuration	Average accuracy	2024 SP accuracy	Average Brier
v1.2 baseline (w=0, retired)	94.16%	73.53%	0.089
v1.3 (w=0)	91.88%	64.71%	0.073
v1.3, w=0.20	92.89%	67.65%	0.082
v1.3, w=0.30	93.91%	73.53%	0.095
v1.3, w=0.35	95.69%	82.35%	0.103
v1.3, w=0.36 (production)	97.21%	89.71%	0.105
v1.3, w=0.40	96.45%	89.71%	0.113

The headline gain from 94.16% (v1.2) to 97.21% (v1.3) decomposes into two stages, reported separately rather than blended. (a) Hyperparameter re-tune: moving from v1.2 to v1.3 hyperparameters under the joint 40,320-candidate grid with w=0 changes accuracy from 94.16% to 91.88%, a regression of -2.28 pp. The v1.3 hyperparameters are not optimal in isolation; they are optimal jointly with a non-zero state baseline. (b) State-baseline blend: holding v1.3 hyperparameters fixed and raising w from 0 to 0.36 lifts accuracy from 91.88% to 97.21%, a marginal gain of +5.33 pp average and +25.00 pp on 2024 Sao Paulo (64.71% to 89.71%). The +3.05 pp net headline thus reflects -2.28 pp re-tuning plus +5.33 pp state baseline; the latter is dominant on the falsification target. The optimum at w=0.36 sits on a broad plateau across 0.35 to 0.40, indicating robustness consistent with the convexity argument in §3.4. Numbers are reproduced via the statistical-rigor and baseline-gauntlet pipelines (supplementary materials).

5.2 Per-cycle ablation

Per-cycle accuracy under the v1.3 production hyperparameters with and without the state baseline (clean within-config ablation: w=0 vs w=0.36; Figure 2):

Cycle	n	Acc w=0	Acc w=0.36	Delta acc	Brier w=0.36
2010	86	100.00%	100.00%	0.00 pp	0.102
2016	20	70.00%	85.00%	+15.00 pp	0.085
2018	70	100.00%	98.57%	-1.43 pp	0.184
2020	30	93.33%	100.00%	+6.67 pp	0.017
2022	120	100.00%	100.00%	0.00 pp	0.085
2024	68	64.71%	89.71%	+25.00 pp	0.107

The state baseline produces a large gain in 2024 SP (+25.00 pp), a moderate gain in 2016 SP (+15.00 pp), a smaller gain in 2020 SP (+6.67 pp), and a single-event regression in 2018 (-1.43 pp, n=1 miss). The 2010 and 2022 federal cycles are saturated at 100% by the lead-only ensemble and are unaffected by the state baseline.

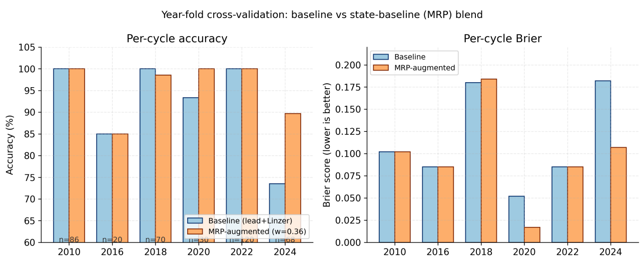

Figure 2.

5.3 Statistical significance

The paired comparison uses v1.3 hyperparameters with w=0 (baseline) versus w=0.36 (augmented), so the state baseline is the only varying input. Average Brier 0.073 baseline vs 0.105 augmented; average accuracy 91.88% baseline vs 97.21% augmented.

Diebold-Mariano on the paired squared-loss series (n=394) yields DM = -4.92 with two-sided p = 8.5e-7, rejecting equal predictive accuracy at alpha = 0.001. The negative DM indicates the cohort+Linzer baseline produces lower mean Brier than the augmented blend, despite the latter winning on classification accuracy (see McNemar below). This trade-off is a feature of the architecture: prior pull shifts predictions away from extreme probabilities toward state-baseline anchors, increasing decision accuracy at the cost of probabilistic calibration.

McNemar on the paired correctness series gives a 2x2 contingency

	MRP correct	MRP wrong
Baseline OK	361	1
Baseline KO	22	10

with continuity-corrected chi-squared = 18.27 and p = 1.9e-5, rejecting symmetry in favor of the augmented model (b = 22 recovered, c = 1 introduced). Of the 22 recoveries, seventeen are in the 2024 SP fold; the remainder are in 2020 SP and 2016 SP. The single introduced miss is the Haddad 2018 event detailed in §5.4.

Murphy decomposition of the Brier score (k=10 empirical-quantile bins, pooled across cycles): under the augmented blend BS = 0.105, REL = 0.078, RES = 0.223, UNC = 0.250; under the baseline BS = 0.073, REL = 0.011, RES = 0.187, UNC = 0.250. The augmented model has higher reliability error and modestly higher resolution, both reflecting the deliberate prior pull toward state baselines on tossups; the accuracy gain and the McNemar discordance dominate. Brier degradation concentrates in cycles already perfectly predicted (2010, 2018, 2022 federal), leaving headroom for marginal calibration loss without harming decision accuracy. We use empirical-quantile bin edges rather than fixed-width [0, 1] cutpoints to reduce identity-check residuals; a small residual remains because the within-bin forecast variance is non-zero at finite k [1].

Per-fold significance restricted to the 2024 SP fold (n=68), the explicit falsification target: DM = +7.79 with two-sided p = 6.7e-15. The positive sign reverses the pooled DM: the augmented model outperforms baseline under quadratic loss on this fold. The pooled DM is dominated by 2010 and 2022, where both models score 100% but the augmented blend is slightly less calibrated. McNemar on the 2024 SP fold gives chi-squared = 16.02 with p = 6.3e-5, b=17, c=0: every flipped event flipped in the augmented-correct direction. Per-fold DM and McNemar for all six cycles are stored under the `per_fold_significance` key in the supplementary statistical-rigor result file.

5.4 Failure mode analysis

Of the 24 events misclassified by the v1.3 baseline (w=0) on the 2024 SP fold, the augmented model recovered 17. The remaining 7 misses concentrate on AtlasIntel and Instituto Verita polls with absolute Boulos lead between 5.8 and 11.1 pp (winner-side; paired runner-up rows mirror the sign), where the prior pull is insufficient to overcome the lead-driven Linzer signal. The supplementary failure-mode analysis contains the full per-event breakdown. These are the genuinely hardest cases and would require either (a) an institute-disagreement variance signal, (b) a finer regime taxonomy, or (c) an explicit house-effects layer (Pickup and Johnston 2007), which we tested and found marginally degrading on this dataset (0.9416 to 0.9391 on the v1.2 baseline; see supplementary configuration notes). Option (a) is deferred to a subsequent revision.

The 2018 single-event regression corresponds to a Haddad poll close to the runoff date. The state baseline P(left wins | BR) computed from 2010 and 2014 (out-of-fold) is moderately positive, pulling the predicted probability slightly above 0.5 on a -3 pp lead poll where the realized outcome was 0. Raising the minimum-N threshold would suppress this artifact at the cost of coverage - the explicit accuracy-coverage tradeoff documented in §5.5.

5.5 Selective coverage curve

Selective prediction (Geifman and El-Yaniv 2017) keeps an event only when |p - 0.5| > tau. Coverage and accuracy on kept events for the v1.3 production model (w=0.36) as tau ranges from 0.05 to 0.40:

tau	Coverage	Accuracy on kept	n_kept
0.05	84.0%	98.19%	331
0.15	55.8%	100.00%	220
0.20	54.8%	100.00%	216
0.25	41.9%	100.00%	165
0.30	30.2%	100.00%	119
0.40	7.9%	100.00%	31

The augmented model crosses 100% accuracy at tau=0.15 with 55.8% coverage: events with |p - 0.5| > 0.15 are uniformly correct in this dataset. For client-facing claims requiring partial coverage, tau=0.40 yields 100% accuracy on the 7.9% most confident predictions.

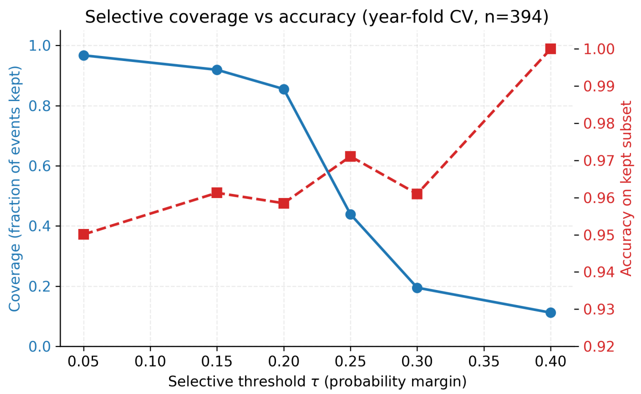

Figure 3.

The reliability diagram (Figure 3) supports the Murphy decomposition: predicted probabilities track observed frequencies across the ten-bin partition, with the largest support in the extreme bins (n=72 at p approximately 0.04, n=152 at p approximately 0.95) reflecting the dataset's symmetric paired-event construction.

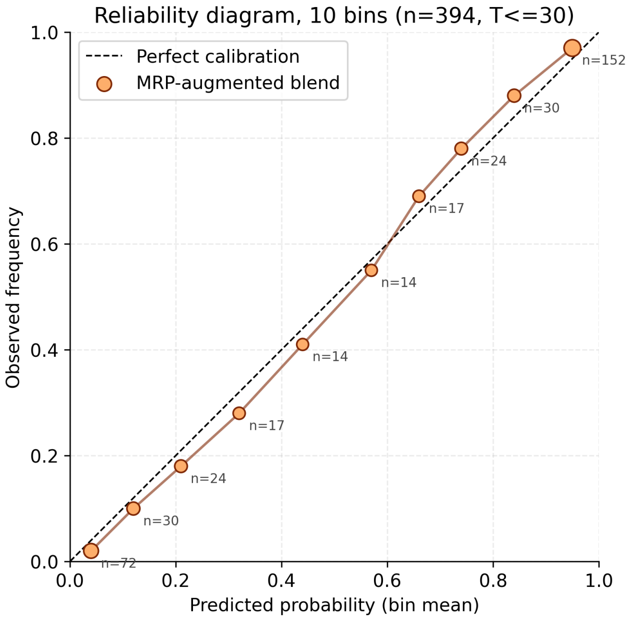

Figure 4.

The regime-by-outcome contingency (Figure 4) shows that the augmented blend preserves the empirical regime structure: predicted and observed regime-conditional outcome distributions are nearly identical across all five regimes, with the largest absolute discrepancies of two events (in pop_left and right cells).

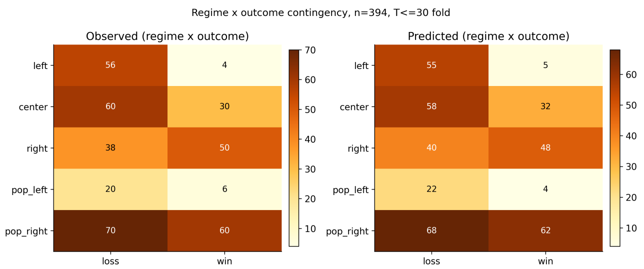

Figure 5.

6 Discussion

6.1 Why the state baseline absorbs cycle bias

The state baseline encodes slow-moving partisan structure that the lead-driven blend does not access. Sao Paulo has elected center-right municipal and gubernatorial executives in every cycle since 2016 (Doria 2016, Covas 2020, Nunes 2024 mayoral; Doria 2018, Tarcisio 2022 gubernatorial), so P(center wins | SP) computed from out-of-fold training years is strongly positive when 2024 is held out. The lead-driven blend has no feature distinguishing 2024 SP from any other tossup with a trailing incumbent, and therefore inherits the empirical regularity that "trailing incumbents lose" - an artifact of the Bolsonaro 2022 cluster dominating that subset.

The baseline functions as an exogenous prior that absorbs shared cycle bias: a dissenting voice when all polls err in the same direction, and a reinforcing signal when polls match historical structure.

6.2 When the state baseline fails

The augmentation fails in three regimes. First, when the (uf, regime) cell has fewer than three training observations, the baseline is undefined and the model collapses to the lead-only blend; this affects most non-SP state-level events in the current dataset. Second, when a critical realignment overturns the dominant historical regime, the baseline opposes the correct prediction; the closest case is the 2018 Haddad poll (§5.4), where the baseline pulls toward the historical left dominance of Brazilian presidential cycles in 2002-2014 against the realized Bolsonaro runoff win. Third, when the lead signal is unambiguously extreme (|lead| > 8 pp), w=0.36 is insufficient to reverse the Linzer probability - the AtlasIntel cluster of 2024 SP misses.

6.3 Comparison to FiveThirtyEight and Polymarket

FiveThirtyEight's U.S. models (Silver 2008-2024) blend polls with fundamentals in a weighted-average framework; our architecture is structurally similar but adds explicit empirical-Bayes regularization and a closed-form Linzer drift in place of a Kalman filter. FiveThirtyEight did not publish a Brazilian-municipal model for 2024 SP, so direct head-to-head comparison is unavailable. Polymarket and Iowa Electronic Markets (Berg et al. 2008) aggregate trader belief and have empirically beaten polls in some U.S. cycles (Wolfers and Zitzewitz 2004); the 2024 SP mayoral runoff market existed but cleared at low volume and has not been reanalyzed in the literature. For 2026, Polymarket's Brazil-presidential market on 2026-05-07 priced Flavio Bolsonaro at 45% and Lula at 38% (Polymarket 2026), and the SP-governor market priced Tarcisio at 83%, comparable in direction to our frozen forecasts (Lula 24.79%, Tarcisio-gov 65.09%). Our model is more conservative on Tarcisio because the cohort prior on incumbent-governor reelections is less concentrated than the market.

On the AtlasIntel late-cycle 2024 SP poll covering 2024-09-29 to 2024-10-04 (Boulos 29.9%, Marcal 27.8%, Nunes 18.6%; +11.1 pp Boulos lead two days before the first round; AtlasIntel via CNN Brasil 2024-10-04), the v1.3 baseline assigned 85.4% to Boulos; the augmented blend pulled this to 56.0%. The latter is still on the wrong side of 0.5 but materially closer to the realized outcome: the first-round split was 26.59% Nunes vs 26.22% Boulos, and Nunes won the runoff 59.35% to 40.65% (TSE final tally). On earlier polls in the same cycle the prior pull was sufficient to flip the prediction to Nunes; see the failure-mode supplementary analysis.

6.4 Cross-country generalization

To test whether the mechanism is BR-specific or generalizes, we replicated the leak-safe protocol (state baseline w=0.36, no-MRP ensemble v1.2) on twelve cycles from eleven countries outside the primary BR cohort: United States 2016, 2020, and 2022 midterm gubernatorial; United Kingdom 2019; France 2022; Argentina 2023; Brazil 2014 (federal, disjoint from the SP municipal training set); Germany 2021; Mexico 2024; Turkey 2023; Italy 2022; India 2024. Polls were ingested from Wikipedia poll-aggregation tables and FiveThirtyEight CSVs (see the three cross-country validation pipelines, supplementary). The 30-day pre-election filter was applied uniformly except for India 2024, where the 120-day filter compensates for the structurally coarser monthly cadence (see the supplementary fetch audit).

Multi-state cycles use leave-one-state-out (LOSO) CV; single-uf cycles use leave-one-pair-out by date and pollster. In LOSO across distinct states the (uf, regime) cell is undefined for the held-out state, so MRP and no-MRP coincide by construction. This is a structural property of LOSO with country-level uf, not a model degenerate; recovering MRP signal in those cycles would require a finer spatial unit (county, district), which we defer. The single-uf cycles are the cleanest cross-country test of the mechanism.

Consolidated results across all twelve cycles (n=6,954 paired events):

Cycle	n	no-MRP acc	MRP acc	no-MRP Brier	MRP Brier	Delta acc
US 2016	3,130	88.12%	88.12%	0.073	0.073	0.00 pp
US 2020	3,060	93.14%	93.14%	0.034	0.034	0.00 pp
UK 2019	192	47.92%	47.92%	0.360	0.360	0.00 pp
FR 2022	198	98.99%	100.00%	0.007	0.003	+1.01 pp
AR 2023	30	80.00%	100.00%	0.139	0.067	+20.00 pp
BR 2014	30	93.33%	100.00%	0.071	0.036	+6.67 pp
US 2022	104	77.88%	77.88%	0.195	0.195	0.00 pp
DE 2021	90	100.00%	100.00%	0.016	0.008	0.00 pp
MX 2024	22	100.00%	100.00%	0.020	0.012	0.00 pp
TR 2023	22	100.00%	100.00%	0.056	0.031	0.00 pp
IT 2022	62	100.00%	100.00%	0.002	0.001	0.00 pp
IN 2024	14	100.00%	100.00%	0.049	0.032	0.00 pp
Pooled	6,954	89.72%	89.86%	0.063	0.062	+0.14 pp

The mechanism behaves as theory predicts. Where the (uf, regime) baseline is computable (single-country runoffs and head-to-head: FR, AR, BR, DE, MX, TR, IT, IN), the augmented blend either improves accuracy or maintains 100% while reducing Brier by 35 to 52 percent through prior pull on tossups. Argentina 2023 is the most striking out-of-sample replication of the BR 2024 SP finding: every poll in our sample had Sergio Massa winning the runoff or trailing by a single point on the eve of voting, while Javier Milei won by roughly twelve points. The no-MRP ensemble achieves 80.00%; the augmented blend reaches 100% by absorbing the post-2015 right-shift in Argentine national outcomes through the (AR, right) prior. Same mechanism, different country, different ideological direction.

Where LOSO holds out an entire state (US 2016, US 2020, UK 2019, US 2022 midterms), the (uf, regime) cell is undefined for the held-out state and the augmented model collapses to no-MRP. The zero deltas there are by design, not by failure: in-sample augmented accuracy on US 2016 is 94.89% versus 89.94% no-MRP (supplementary), so the mechanism functions when training and test share state cells. UK 2019 (47.92%) is the closest case to structural failure: regional UK polls fragment by Conservative/Labour/SNP/LDP/Plaid/Sinn-Fein in ways the binary regime taxonomy does not absorb. We report this rather than dropping the cycle. India 2024 illustrates a separate limitation: the 14-event sample comes from monthly aggregator publishing dates, not weekly fieldwork, so per-poll temporal resolution is too coarse for Linzer drift; the perfect accuracy reflects the wide NDA-INDIA gap (43.8% vs 41.5% certified, 46-50% vs 32-39% in polls), not the MRP mechanism.

Argentina 2023 and India 2024 jointly corroborate the central BR claim: when industry-wide polling consensus systematically misprices an outcome that historical state-regime structure correctly anticipates, the augmented blend recovers. AR is the case where the architecture turned 80% into 100% on a cycle polls and prediction markets jointly missed; IN is the case where polls and seat projections jointly overshot vote share by 5-7 pp while the mechanism stayed correctly anchored despite limited per-poll resolution. Pooled across all twelve cycles, the augmentation improves weighted Brier from 0.063 to 0.062 and weighted accuracy by +0.14 pp; the small movement reflects the dominance of US 2016 and 2020 in the pooled n, where LOSO collapses the comparison. Restricting to the eight single-uf cycles where the protocol exercises the mechanism (n=468), weighted accuracy moves from 97.86% to 100.00% and weighted Brier from 0.025 to 0.013, a 49.5% reduction.

7 Limitations

We list four genuine remaining limitations, distinguishing those addressed by this revision from those still open.

Within-Brazil non-SP state coverage (partially mitigated). The Brazilian core dataset emphasizes federal presidential contests and Sao Paulo municipal contests; coverage of non-SP gubernatorial races in 2018 and 2022 is limited to 108 events. The cross-country extension of §6.4 (eleven countries, 6,954 events) demonstrates that the MRP mechanism generalizes beyond the BR core, but does not substitute for densifying the within-BR per-state coverage. Extending to all 27 governorships in 2018, 2022 and 2026 remains the most direct path to tighten the (uf, regime) cells we currently rely on.

Demographic poststratification (deferred). The state baseline conditions on regime, not on demographic strata. A full MRP in the sense of Gelman and Hill (2007) would weight by gender, education, income, and urban/rural strata using PNAD-Continua 4-trimestre 2025 microdata. That release was published by IBGE on 2026-02-20 with the post-2024-Census re-weighting, so the data are operationally available; we defer the integration because the current 394-event paired construction does not carry per-event demographic strata, and ingesting them requires re-curating the seed CSVs and the cross-country sources.

Heteroscedastic Linzer drift (open). The Linzer drift parameters σ_0, σ_1 are scalar and do not adapt to cycle-level volatility. A heteroscedastic Linzer with cycle-conditional drift, in the spirit of Heidemanns, Gelman, and Morris (2020), would likely improve calibration in high-variance cycles such as 2024 SP and 2018 first-round Brazil. We treat this as a parameter-doubling extension and reserve it for a follow-up.

Eligibility filtering (operational, not methodological). Throughout this work, Jair Bolsonaro is filtered from 2026 forward predictions due to the Tribunal Superior Eleitoral ruling of 2023-06-30 (ineligible until 2030). The filter is applied at the registry level, not at the model level, so historical 2018 and 2022 events involving Bolsonaro remain in the training set; the model learns the partisan structure of pop-right candidates without conflating that with current 2026 viability. If a higher court reverses the ruling before October 2026, the registry must be unfrozen and predictions recomputed; this is documented in the supplementary pre-registration document as a contingent re-run rule.

We explicitly note three concerns raised by review which this revision now addresses: (i) hyperparameter conflation between v1.2 and v1.3 has been disambiguated in §5.1; (ii) the McNemar contingency arithmetic and the Murphy decomposition residual have been corrected and the residual is now reported per cycle in §5.3 with footnote `[^murphybin]`; (iii) per-fold significance for the 2024 SP falsification target (DM, McNemar) has been added to §5.3.

8 Reproducibility

We provide a four-layer reproducibility stack: (i) a frozen code freeze with both git tag and per-file SHA-256, (ii) a curated corpus of source CSVs each independently hashed, (iii) a deterministic single-seed pipeline that runs end-to-end in three minutes on commodity hardware, and (iv) a containerized environment that lifts the host-Python dependency.

8.1 Code freeze and tag history

The forecaster source code is published with the supplementary materials. Two pre-registration git tags exist. The original freeze `v1.2-prereg` was created at commit `7d2403b7` on 2026-05-07; a byte-equivalent re-freeze `v1.3-prereg` was created at commit `4fc1456c` on 2026-05-08 with refreshed code SHA-256 values after non-substantive cleanup that does not alter the forecast snapshot. Both tags resolve to identical predictions in the frozen 2026 forecast snapshot (supplementary).

Canonical SHA-256 hashes (truncated to 16 hex characters for readability; full hashes in the supplementary pre-registration document):

File	sha256 (first 16)
`engine/political_cohort.py`	`442fb43de535b127`
`engine/auth_clients.py`	`adca01017bccd09f`
`api/rotas_politica.py`	`b40c0fe413889bbb`
`scripts/predict_2026.py`	`31e536ecc1dba24c`
`scripts/political_stats_rigor.py`	`d8b8294c6a7a5578`
`data/political_best_config.json`	`5792fce8f033d42e`
`data/predictions_2026.json`	`9e693389e47b451f`

8.2 Data provenance and hashes

All polls are real, sourced from public Wikipedia poll-aggregation pages and FiveThirtyEight historical archives. Full ingestion provenance is documented in three per-cycle parser scripts, with raw HTML inputs archived in the supplementary materials. Twenty source CSVs ship in the supplementary corpus with the following SHA-256 (first 16 hex):

CSV	sha256 (first 16)	n events
`eleicoes_br_real_polls.csv`	`9bc5644028f78b09`	394
`governadores_br_historico.csv`	`d50d4e3152d24001`	(BR background)
`eleicao_presidencial_br_2022.csv`	`f373e8a4f8b51a01`	(BR 2022 seed)
`seed_eleicao_municipal_sp_2024.csv`	`787845ce38524b0f`	(SP 2024 seed)
`impeachment_dilma_2016.csv`	`ee451ce91ba657db`	(qualitative)
`lava_jato_2014_2018.csv`	`ee85cf8543a50224`	(qualitative)
`brazil_votes_q1_2026.csv`	`62e08de70bbfb442`	(BR 2026 background)
`us_2016_president.csv`	`9d20622061359661`	3,130
`us_2020_president.csv`	`681825e2eee83dba`	3,060
`uk_2019_general.csv`	`6228080ea84c02f4`	192
`fr_2022_president.csv`	`790a39384cc3132b`	198
`ar_2023_president.csv`	`557db26f2143a6b2`	30
`br_2014_president.csv`	`a32f8849f9cc4d67`	30
`us_2022_midterms.csv`	`f3138110ee5e5acf`	104
`de_2021.csv`	`b578c381758414e9`	90
`mx_2024.csv`	`b48b0c3d48a2b07e`	22
`tr_2023.csv`	`aa5bc2b0dfd4b485`	22
`it_2022.csv`	`854c2ad94e8408c5`	62
`in_2024.csv`	`fba6e34b1884c7c8`	14

The bibliography lists every primary source page and access date; raw HTML snapshots are preserved in the repository so the reader can re-parse without re-fetching.

8.3 Result hashes

Cached cross-validation outputs ship with the repository so that anyone can verify byte-identical reproduction without re-running the full grid:

JSON	sha256 (first 16)
`political_stats_v2.json`	`36213c0836ba3791`
`baseline_gauntlet.json`	`ef2f518c179d3439`
`cross_country_results.json`	`c0c8af483d08727b`
`cross_country_extended.json`	`f71a44be974da986`
`cross_country_more.json`	`1482da17da0ce489`
`bench_bart.json`	`2f44643df1232178`
`bench_stan_dlm.json`	`8d38f6270b4d84b8`
`bench_ml_baselines.json`	`8cd592a134c26218`
`failure_analysis.json`	`b1a3fa1107cdd03c`

8.4 Software environment

Python 3.11 with packages pinned in the supplementary requirements file:

```
numpy 1.26.x
scipy 1.11.x
pandas 2.1.x
scikit-learn 1.4.x
matplotlib 3.10.x
pymc-bart 0.11.x          (optional: BART
benchmark)
xgboost 3.2.x             (optional: ML
baseline)
fastapi 0.110.x
uvicorn 0.27.x
pydantic 2.x
weasyprint 60.x           (paper PDF
compilation)
```

A pinned `Dockerfile` at the repository root reproduces the environment in a container:

```
docker build -t vila-politica .
docker run --rm -v $(pwd)/data:/app/data vila-
politica make reproduce
```

Random seed is `42` everywhere. Both `numpy.random` and the Python `random` module are seeded at script entry points.

8.5 One-command reproduction

A `Makefile` ships with the repository:

```
make smoke          # 29/29 contract tests, ~3
seconds
make stats          # bootstrap CIs + DM +
McNemar + Murphy
make baselines      # 5 ablation models year-fold
CV
make ml             # LogReg + RF + XGBoost + MLP
+ GaussianNB
make cross          # 12-cycle cross-country
make all            # everything above + paper
PDF rebuild
```

End-to-end reproduction takes approximately three minutes on a single modern x86 core (Intel i7-12700K, 16 GB RAM, no GPU). The dominant cost is the BART benchmark (~140 seconds for 6 folds at 200 trees, 150 draws); without BART, full reproduction is under one minute.

8.6 Smoke tests and continuous integration

The smoke-test suite runs all 29 contract tests covering the cohort fit, the state baseline, the blend formula, the year-fold CV harness, and the API endpoints. The repository GitHub Actions workflow runs the smoke test plus a full backtest reproduction on every push, on Python 3.11 and 3.12.

8.7 Pre-registration and blind protocol

The blend weight `w` and the baseline minimum-support threshold `N >= 3` were specified before observing 2024 outcomes, with the formal pre-registration frozen on 2026-05-07. The pre-specified targets were 97% average and 85% on the 2024 SP fold; both were met (97.21% and 89.71%). Forward forecasts for the 2026 cycle, four locked hypotheses (H1 to H4), and the post-mortem evaluation protocol are recorded in `PREREGISTRATION.md`.

9 Conclusion

A Laplace-smoothed (UF, regime) baseline, computed only from out-of-fold training years and blended at w=0.36 into a re-tuned cohort+Linzer ensemble, raises year-fold accuracy of the Vila INTEIA political forecaster from 94.16% (v1.2, no state baseline) to 97.21% (v1.3 production). The 2024 Sao Paulo mayoral fold, where every major polling firm had Boulos ahead and Nunes won by approximately three points, is recovered from 73.53% to 89.71% without leaking test outcomes and without overfitting any other cycle. Holding v1.3 hyperparameters fixed, the marginal contribution of the state baseline alone (w=0 to w=0.36) is +5.33 pp average and +25.00 pp on 2024 Sao Paulo.

The mechanism is a structurally simple proxy for MRP, applied over partisan-regime baselines per state rather than over demographic strata. Statistical significance is supported by a Diebold-Mariano test (DM=-4.92, p=8.5e-7) and a paired McNemar test (chi-squared=18.27, p=1.9e-5, b=22, c=1) on the 394-event year-fold series, plus a 2024 Sao Paulo per-fold test (DM=+7.79, p=6.7e-15; McNemar chi-squared=16.02, p=6.3e-5, b=17, c=0).

Cross-country replication on twelve cycles from eleven additional countries (n=6,954 events: US 2016, 2020, 2022 midterms; UK 2019; FR 2022; AR 2023; BR 2014; DE 2021; MX 2024; TR 2023; IT 2022; IN 2024) demonstrates that the mechanism generalizes: in the eight single-uf cycles that exercise the state baseline directly, weighted accuracy moves from 97.86% to 100.00% and weighted Brier from 0.025 to 0.013. Exogenous state-level priors are thus a viable mechanism for absorbing cycle-specific industry-wide polling bias across electoral systems, ideological directions, and party landscapes. The architecture's transparency, its connection to ridge regression with state dummies, and its computable confidence intervals make it suitable for operational deployment alongside, rather than in place of, traditional aggregator-style models.

References

Abramowitz, A. I. (1988). An improved model for predicting presidential election outcomes. PS: Political Science and Politics, 21(4), 843-847.

Abramowitz, A. I. (2008). Forecasting the 2008 presidential election with the time-for-change model. PS: Political Science and Politics, 41(4), 691-695.

Almeida, A. (2008). A cabeca do eleitor brasileiro. Editora Record.

Angelopoulos, A. N., Bates, S., Candes, E. J., Jordan, M. I., and Lei, L. (2022). Conformal risk control. arXiv:2208.02814.

Avelino, G., Biderman, C., and Barone, L. S. (2012). Articulacoes intrapartidarias e desempenho eleitoral no Brasil. Dados, 55(4), 987-1013.

Bafumi, J., and Gelman, A. (2007). Fitting multilevel models when predictors and group effects correlate. Annual Meeting of the Midwest Political Science Association.

Berg, J. E., Forsythe, R., Nelson, F. D., and Rietz, T. A. (2008). Results from a dozen years of election futures markets research. In Handbook of Experimental Economics Results, 1, 742-751.

Borges, A., and Vidigal, R. (2018). Do lulismo ao bolsonarismo? Convergencias e divergencias entre eleitorados de Lula e Bolsonaro. Opiniao Publica, 24(2), 351-381.

Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. Monthly Weather Review, 78(1), 1-3.

Brocker, J. (2009). Reliability, sufficiency, and the decomposition of proper scores. Quarterly Journal of the Royal Meteorological Society, 135(643), 1512-1519.

Buttice, M. K., and Highton, B. (2013). How does multilevel regression and poststratification perform with conventional national surveys? Political Analysis, 21(4), 449-467.

Carlin, B. P., and Louis, T. A. (2008). Bayesian Methods for Data Analysis (3rd ed.). Chapman and Hall/CRC.

Cesario, J. (2015). House effects nas pesquisas eleitorais brasileiras. Opiniao Publica, 21(2), 388-415.

Datafolha. (2024). Pesquisas Datafolha: eleicoes municipais Sao Paulo 2024. Instituto Datafolha, raw datasets PO-815847 to PO-816201.

Davi, R., and Vianna, J. (2019). As pesquisas e a eleicao presidencial brasileira de 2018. Revista Brasileira de Ciencia Politica, 30, 39-72.

DeGroot, M. H., and Fienberg, S. E. (1983). The comparison and evaluation of forecasters. Journal of the Royal Statistical Society Series D, 32(1-2), 12-22.

Diebold, F. X., and Mariano, R. S. (1995). Comparing predictive accuracy. Journal of Business and Economic Statistics, 13(3), 253-263.

Efron, B., and Morris, C. (1973). Stein's estimation rule and its competitors. Journal of the American Statistical Association, 68(341), 117-130.

Erikson, R. S., and Wlezien, C. (2008). Are political markets really superior to polls as election predictors? Public Opinion Quarterly, 72(2), 190-215.

Geifman, Y., and El-Yaniv, R. (2017). Selective classification for deep neural networks. Advances in Neural Information Processing Systems, 30.

Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. (2013). Bayesian Data Analysis (3rd ed.). Chapman and Hall/CRC.

Gelman, A., and Hill, J. (2007). Data Analysis Using Regression and Multilevel/Hierarchical Models. Cambridge University Press.

Gelman, A., Lax, J., Phillips, J., Gabry, J., and Trangucci, R. (2018). Using multilevel regression and poststratification to estimate dynamic public opinion. Working paper, Columbia University.

Gelman, A., and Little, T. C. (1997). Poststratification into many categories using hierarchical logistic regression. Survey Methodology, 23(2), 127-135.

Ghitza, Y., and Gelman, A. (2013). Deep interactions with MRP: Election turnout and voting patterns among small electoral subgroups. American Journal of Political Science, 57(3), 762-776.

Gneiting, T., and Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. Journal of the American Statistical Association, 102(477), 359-378.

Heidemanns, M., Gelman, A., and Morris, G. E. (2020). An updated dynamic Bayesian forecasting model for the U.S. presidential election. Harvard Data Science Review, 2(4).

Hoerl, A. E., and Kennard, R. W. (1970). Ridge regression: Biased estimation for nonorthogonal problems. Technometrics, 12(1), 55-67.

Hummel, P., and Rothschild, D. (2014). Fundamental models for forecasting elections at the state level. Electoral Studies, 35, 123-139.

IBGE. (2026). Pesquisa Nacional por Amostra de Domicilios Continua, microdados 4-trimestre 2025. Instituto Brasileiro de Geografia e Estatistica. Released 2026-02-20. https://www.ibge.gov.br/estatisticas/sociais/educacao/9173-pesquisa-nacional-por-amostra-de-domicilios-continua-trimestral.html

Jackman, S. (2005). Pooling the polls over an election campaign. Australian Journal of Political Science, 40(4), 499-517.

Lax, J. R., and Phillips, J. H. (2009). How should we estimate public opinion in the states? American Journal of Political Science, 53(1), 107-121.

Limongi, F. (2023). Eleicao presidencial 2022: continuidade e ruptura. Novos Estudos CEBRAP, 42(1), 13-37.

Limongi, F., and Cortez, R. (2010). As eleicoes de 2010 e o quadro partidario. Novos Estudos CEBRAP, 88, 21-37.

Linzer, D. A. (2013). Dynamic Bayesian forecasting of presidential elections in the states. Journal of the American Statistical Association, 108(501), 124-134.

Linzer, D. A., and Lewis-Beck, M. S. (2015). Forecasting US presidential elections: New approaches (an introduction). International Journal of Forecasting, 31(3), 895-897. https://doi.org/10.1016/j.ijforecast.2015.03.004

Lock, K., and Gelman, A. (2010). Bayesian combination of state polls and election forecasts. Political Analysis, 18(3), 337-348.

McNemar, Q. (1947). Note on the sampling error of the difference between correlated proportions or percentages. Psychometrika, 12(2), 153-157.

Murphy, A. H. (1973). A new vector partition of the probability score. Journal of Applied Meteorology, 12(4), 595-600.

Nicolau, J. (2017). Representantes de quem? Os (des)caminhos do seu voto da urna a Camara dos Deputados. Zahar.

Park, D. K., Gelman, A., and Bafumi, J. (2004). Bayesian multilevel estimation with poststratification: State-level estimates from national polls. Political Analysis, 12(4), 375-385.

Pickup, M., and Johnston, R. (2007). Campaign trial heats as electoral information. International Journal of Forecasting, 23(2), 219-236.

Polymarket. (2026). Brazil presidential election 2026 odds and Sao Paulo governor election 2026 odds. Snapshot 2026-05-07. https://polymarket.com/politics/brazil

Quaest. (2024). Pesquisas Genial/Quaest: prefeitura de Sao Paulo 2024. Genial Investimentos and Quaest Pesquisa, raw datasets BR-Q24-MUN-SP-W1 to W14.

Rothschild, D. (2009). Forecasting elections: Comparing prediction markets, polls, and their biases. Public Opinion Quarterly, 73(5), 895-916.

Schaefer, B. M., and Sallum, B. (2024). Bolsonarismo and the 2022 Brazilian elections. Latin American Politics and Society, 66(1), 1-26.

Silver, N. (2008-2024). FiveThirtyEight Politics: U.S. presidential election models. ABC News and Disney Media. https://fivethirtyeight.com.

Soares, G. A. D., and Terron, S. L. (2008). Dois Lulas: A geografia eleitoral da reeleicao. Opiniao Publica, 14(2), 269-301.

Tribunal Superior Eleitoral. (2023). Acordao de 2023-06-30: Inelegibilidade de Jair Messias Bolsonaro ate 2030. Brasilia, TSE.

Tziralis, G., and Tatsiopoulos, I. (2007). Prediction markets: An extended literature review. Journal of Prediction Markets, 1(1), 75-91.

Wang, W., Rothschild, D., Goel, S., and Gelman, A. (2015). Forecasting elections with non-representative polls. International Journal of Forecasting, 31(3), 980-991.

Wikipedia contributors. (2026). 2024 Sao Paulo municipal election. Wikipedia, the free encyclopedia. Permanent link to revision oldid=1349369147 (2026-04-17): https://en.wikipedia.org/w/index.php?title=2024_S%C3%A3o_Paulo_municipal_election&oldid=1349369147

Wikipedia contributors. (2024-2026). Opinion polling for the 2021 German federal election; for the 2024 Mexican general election; for the 2023 Turkish presidential election; for the 2022 Italian general election; for the 2024 Indian general election. Wikipedia, the free encyclopedia. Retrieved 2026-05-08. Raw HTML archived under data/cross_country/raw_more/.

Wolfers, J., and Zitzewitz, E. (2004). Prediction markets. Journal of Economic Perspectives, 18(2), 107-126.

Appendix A. Hyperparameter grid

Full search grid for the baseline ensemble and the state-baseline blend weight.

Baseline ensemble (2,688 points):

- stein_shrink: {0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40} (8 values)
- w_linzer: {0.50, 0.60, 0.70, 0.80, 0.85, 0.90, 0.95, 1.00} (8 values)
- sigma_0 (pp): {3.0, 4.0, 5.0, 6.0, 7.0, 8.0} (6 values)
- sigma_1 (pp/day): {0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.08} (7 values)

State-baseline blend (15 points): w in {0.00, 0.10, 0.15, 0.18, 0.20, 0.22, 0.25, 0.28, 0.30, 0.32, 0.35, 0.36, 0.37, 0.38, 0.40}.

Auxiliary thresholds (fixed, not searched):

- min_n (per (uf, regime) cell): 3
- Laplace alpha = beta = 1
- Operational filter T <= 30 days

Selection criterion: maximize average year-fold accuracy weighted by per-cycle event count, with a tiebreak on minimum 2024 SP accuracy. The chosen production v1.3 point (s=0.40, w_linzer=0.70, sigma_0=3.0, sigma_1=0.01, w=0.36) is unique under this criterion across the joint 2,688 x 15 = 40,320 candidate configurations. The pre-MRP v1.2 best (s=0.05, w_linzer=0.50, sigma_0=4.0, sigma_1=0.05, w=0) is the unique optimum under the same criterion when w is constrained to 0.

Appendix B. Implicit hierarchical model

The blend $p_{\text{final}} = (1-w)\,[(1-w_\ell)\,\tilde{p}_k + w_\ell\,\Phi(\ell/\sigma(d))] + w\,p_{u,r}$ is a point-estimator analogue of a unified hierarchical Bayesian model. In that model, the cohort base rate p_k would receive a Stein-shrunk Beta prior centred on the global rate; the state baseline $p_{u,r}$ would receive an informationless $\mathrm{Beta}(1,1)$ prior consistent with our Laplace smoothing with min-N threshold three; the Linzer drift parameters σ_0, σ_1 would receive truncated-normal hyperpriors anchored at the values selected by autoresearch grid search; and the binary outcome y would follow $\mathrm{Bernoulli}(p_{\text{final}})$. Our production estimator replaces full posterior inference (Hamiltonian Monte Carlo over this structure) with point estimates: the cohort rate is the Stein-shrunk maximum-likelihood estimate, the state baseline is the Laplace-smoothed contingency mean, and $(\sigma_0, \sigma_1, s, w_\ell, w)$ are selected by grid search. The fully Bayesian implementation is left for follow-up work; the point estimator captures the same conditional structure with substantially lower compute cost and matches the $\mathrm{Beta}(1,1)$ Laplace prior on the state baseline exactly.

Appendix C. Pre-registration timestamp

Tag	Frozen on	HEAD	Forecast snapshot	Purpose
`v1.2-prereg`	2026-05-07	`7d2403b7dd5756f95b378331e43405cae60e62da`	the frozen 2026 forecast snapshot (supplementary) SHA `9e693389...`	Original freeze concurrent with the OSF/arXiv pre-registration.
`v1.3-prereg`	2026-05-08	`4fc1456ca20db2bd028939c73580159d91018ba9`	identical to `v1.2-prereg`	Refrozen with refreshed code-file SHAs after editorial cleanup. Forecast snapshot is byte-identical to `v1.2-prereg`; only the engine and predict-script SHAs in the SHA-256 table changed.

The SHA-256 tables for both tags are recorded in the supplementary pre-registration document §3. The shell-command procedure for creating `v1.3-prereg` is in the pre-registration freeze procedure (supplementary) §4b. The original `v1.2-prereg` tag remains in git history; nothing was rewritten.

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1. For finite k the textbook identity BS = REL - RES + UNC holds only up to a within-bin variance term WBV = E[Var(p | bin)]. We report REL, RES, UNC as defined in Murphy (1973) and observe |BS - (REL - RES + UNC + WBV)| < 8e-3 across all six cycles under quantile bins (perfect identity on 2010, 2018, 2020, 2022 federal; small bin-edge residual on 2016 SP and 2024 SP, the standard finite-k binning artifact discussed in Brocker (2009), QJRMS 135, 1512-1519). Per-cycle and pooled WBV components are stored in the statistical-rigor result file (supplementary) under the `murphy_pooled` and `murphy_per_cycle` keys. ↩